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Chapter 8

The Hecke Type Zeta-functions

In this chapter we consider the special case of the Main Formula (Chapter 7) in which the Dirichlet series $\phi(s), \psi(s)$ satisfy the **Hecke type functional equation** (8.1). The modular relation (8.6) includes the Bochner modular relation (cf. Theorem 3.4 etc.), the Fourier-Bessel expansion (Chapter 4), the Ewald expansion (Chapter 5) which includes the Bochner-Chandrasekharan formula (§5.1.2), the Riesz sum (Chapter 6). We shall state special cases of (8.7), which is to work as a database. Examples of non-Hecke type were studied in [BeI; BeII; BeIII; BeIV], and also in [KTT1; KTT2] many of which may be reduced to Hecke type and the corresponding results can be readily guessed from the database.

8.1 Statement of the formula

Given two Dirichlet series, we assume the functional equation of Hecke type

$$\chi(s) = \begin{cases} \Gamma(s)\,\varphi(s), & \mathrm{Re}(s) > \sigma_\varphi \\ \Gamma(r-s)\,\psi(r-s), & \mathrm{Re}(s) < r - \sigma_\psi, \end{cases} \tag{8.1}$$

or

$$\chi(s) = \begin{cases} \Gamma\left(s \,\middle|\, \begin{matrix} - & ; & - \\ (0,1) & ; & - \end{matrix}\right)\varphi(s), & \mathrm{Re}(s) > \sigma_\varphi \\ \Gamma\left(r-s \,\middle|\, \begin{matrix} - & ; & - \\ (0,1) & ; & - \end{matrix}\right)\psi(r-s), & \mathrm{Re}(s) < r - \sigma_\psi. \end{cases} \tag{8.2}$$

By Theorem 7.1, we have the following modular relation:

$$\begin{aligned}
&\sum_{k=1}^{\infty} \alpha_k H\Bigg(z\lambda_k \;\Bigg|\; (\Delta - s) \oplus \begin{pmatrix} - & ; & - \\ (0,1) & ; & - \end{pmatrix} \Bigg) \\
&= z^{-r} \sum_{k=1}^{\infty} \beta_k H\Bigg(\frac{\mu_k}{z} \;\Bigg|\; (\Delta^* - (r-s)) \oplus \begin{pmatrix} - & ; & - \\ (0,1) & ; & - \end{pmatrix} \Bigg) \qquad (8.3) \\
&\quad + \sum_{k=1}^{L} \operatorname{Res}\Big(\Gamma(w - s \,|\, \Delta)\, \chi(w)\, z^{-w}, w = s_k \Big),
\end{aligned}$$

which can be rewritten as

$$\begin{aligned}
&\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s} H\Bigg(z\lambda_k \;\Bigg|\; \Delta \oplus \begin{pmatrix} - & ; & - \\ (s,1) & ; & - \end{pmatrix} \Bigg) \\
&= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}} H\Bigg(\frac{\mu_k}{z} \;\Bigg|\; \Delta^* \oplus \begin{pmatrix} - & ; & - \\ (r-s,1) & ; & - \end{pmatrix} \Bigg) \\
&\quad + \sum_{k=1}^{L} \operatorname{Res}\Big(\Gamma(w - s \,|\, \Delta)\, \chi(w)\, z^{s-w}, w = s_k \Big).
\end{aligned}$$

In the traditional notation, if

$$\Delta = \begin{pmatrix} \{(1-a_j, A_j)\}_{j=1}^{n}; & \{(a_j, A_j)\}_{j=n+1}^{p} \\ \{(b_j, B_j)\}_{j=1}^{m} & ;\{(1-b_j, B_j)\}_{j=m+1}^{q} \end{pmatrix} \in \Omega \qquad (8.4)$$

with its associated Gamma factor

$$\Gamma(s \,|\, \Delta) = \frac{\displaystyle\prod_{j=1}^{m} \Gamma\big(b_j + B_j s\big) \prod_{j=1}^{n} \Gamma\big(a_j - A_j s\big)}{\displaystyle\prod_{j=n+1}^{p} \Gamma\big(a_j + A_j s\big) \prod_{j=m+1}^{q} \Gamma\big(b_j - B_j s\big)}, \qquad (8.5)$$

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we have

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s} H_{p,q+1}^{m+1,n}\left(z\lambda_k \,\middle|\, \begin{matrix} \{(1-a_j, A_j)\}_{j=1}^{n}, \{(a_j, A_j)\}_{j=n+1}^{p} \\ (s,1), \{(b_j, B_j)\}_{j=1}^{m}, \{(1-b_j, B_j)\}_{j=m+1}^{q} \end{matrix} \right) \tag{8.6}$$

$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}} H_{q,p+1}^{n+1,m}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} \{(1-b_j, B_j)\}_{j=1}^{m}, \{(b_j, B_j)\}_{j=m+1}^{q} \\ (r-s,1), \{(a_j, A_j)\}_{j=1}^{n}, \{(1-a_j, A_j)\}_{j=n+1}^{p} \end{matrix} \right)$$

$$+ \sum_{k=1}^{L} \mathrm{Res}\Big(\Gamma(w-s \,|\, \Delta)\, \chi(w)\, z^{s-w}, w = s_k \Big).$$

In the special case of $A_j = B_j = 1$, we have

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s} G_{p,q+1}^{m+1,n}\left(z\lambda_k \,\middle|\, \begin{matrix} 1-a_1, \ldots, 1-a_n, a_{n+1}, \ldots, a_p \\ s, b_1, \ldots, b_m, 1-b_{m+1}, \ldots, 1-b_q \end{matrix} \right) \tag{8.7}$$

$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}} G_{q,p+1}^{n+1,m}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-b_1, \ldots, 1-b_m, b_{m+1}, \ldots, b_q \\ r-s, a_1, \ldots, a_n, 1-a_{n+1}, \ldots, 1-a_p \end{matrix} \right)$$

$$+ \sum_{k=1}^{L} \mathrm{Res}\Big(\Gamma(w-s \,|\, \Delta)\, \chi(w)\, z^{s-w}, w = s_k \Big).$$

8.1.1 *The bilateral form*

The special case of (8.7) with

$$m = n,\ p = q,\ a_1 = b_1 - c, \ldots, a_n = b_n - c,\ b_{n+1} = a_{n+1} - c, \ldots, b_p = a_p - c$$

leads to

$$\Delta = \begin{pmatrix} \{(1-b_j+c, 1)\}_{j=1}^{n}; & \{(a_j, 1)\}_{j=n+1}^{p} \\ \{(b_j, 1)\}_{j=1}^{n} & ; \{(1-a_j+c, 1)\}_{j=m+1}^{p} \end{pmatrix}$$

and

$$\Gamma(s \,|\, \Delta) = \frac{\prod_{j=1}^{n} \Gamma(b_j + s) \prod_{j=1}^{n} \Gamma(b_j - c - s)}{\prod_{j=n+1}^{p} \Gamma(a_j + s) \prod_{j=n+1}^{p} \Gamma(a_j - c - s)}.$$

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$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s}\, G^{n+1,n}_{p,p+1}\left(z\lambda_k \,\middle|\, \begin{matrix} 1-b_1+c,\dots,1-b_n+c, a_{n+1},\dots,a_p \\ s, b_1,\dots,b_n, 1-a_{n+1}+c,\dots,1-a_p+c \end{matrix} \right)$$

$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}}\, G^{n+1,n}_{p,p+1}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-b_1,\dots,1-b_n, a_{n+1}-c,\dots,a_p-c \\ r-s, b_1-c,\dots,b_n-c, 1-a_{n+1},\dots,1-a_p \end{matrix} \right)$$

$$+ \sum_{k=1}^{L} \operatorname{Res}\Big(\Gamma(w-s\,|\,\Delta)\,\chi(w)\, z^{s-w}, w=s_k \Big).$$

Since the G-function coefficients on the right-side becomes

$$\frac{\beta_k}{\mu_k^{r-s}} \left(\frac{\mu_k}{z}\right)^{-c} G^{n+1,n}_{p,p+1}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-b_1+c,\dots,1-b_n+c, \\ r-s+c, b_1,\dots,b_n, \\ a_{n+1},\dots,a_p \\ 1-a_{n+1}+c,\dots,1-a_p+c \end{matrix} \right),$$

we have the bilateral formula

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s}\, G^{n+1,n}_{p,p+1}\left(z\lambda_k \,\middle|\, \begin{matrix} 1-b_1+c,\dots,1-b_n+c, a_{n+1},\dots,a_p \\ s, b_1,\dots,b_n, 1-a_{n+1}+c,\dots,1-a_p+c \end{matrix} \right)$$

$$= z^c \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r+c-s}}\, G^{n+1,n}_{p,p+1}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-b_1+c,\dots,1-b_n+c, \\ r-s+c, b_1,\dots,b_n, \\ a_{n+1},\dots,a_p \\ 1-a_{n+1}+c,\dots,1-a_p+c \end{matrix} \right) \tag{8.8}$$

$$+ \sum_{k=1}^{L} \operatorname{Res}\Big(\Gamma(w-s\,|\,\Delta)\,\chi(w)\, z^{s-w}, w=s_k \Big).$$

For each $p \geq 0$, we may write down the formula similar to (4.84) corresponding to n, $\frac{p}{2} \leq n \leq p$.

Convention

In almost what follows we shall state specific form of the general formula (8.7) described in the section titles and we omit this passage. Also in most case $z > 0$ or more generally $\operatorname{Re} z > 0$. It would be instructive to check the validity of various identities among G-functions that occur subsequently.

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8.1.2 *The Bochner modular relation:* $G^{1,0}_{0,1} \leftrightarrow G^{1,0}_{0,1}$

This is the genesis of all subsequent studies on the modular relations and has been fully discussed in general form in Chapter 3 above.

8.2 The Riesz sums or the first J-Bessel expansion: $G^{1,0}_{1,1} \leftrightarrow G^{1,0}_{0,2}$

We have

$$\begin{aligned}
&\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s}\, G^{1,0}_{1,1}\left(z\,\lambda_k \,\middle|\, \begin{matrix} a \\ s \end{matrix} \right) \\
&= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}}\, G^{1,0}_{0,2}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} - \\ r-s, 1-a \end{matrix} \right) \qquad (8.9)\\
&\quad + \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\chi(w)\, z^{s-w}}{\Gamma(a-s+w)}, w = s_k \right) \quad (z>0),
\end{aligned}$$

which by (6.5) and (2.110) amounts to

$$\begin{aligned}
&\frac{z^s}{\Gamma(a-s)} \sum_{\lambda_k < \frac{1}{z}} \alpha_k (1 - z\lambda_k)^{a-s-1} \\
&= z^{\frac{1}{2}(s-r-1+a)} \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{\frac{1}{2}(r-s-1+a)}}\, J_{r-s-1+a}\left(2\sqrt{\frac{\mu_k}{z}} \right) \qquad (8.10)\\
&\quad + \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\chi(w)\, z^{s-w}}{\Gamma(a-s+w)}, w = s_k \right) \quad (z>0).
\end{aligned}$$

The special case of (8.10) with $a = 1$, after slight changes of notation gives rise to the formula for the Riesz sum of order κ of the Idealfunktion of an imaginary quadratic field due to Landau [LanII], Hecke [Hecl], Chandrasekharan and Narasimhan [ChN2, Lemma 5] et al:

Theorem 8.1

$$\frac{1}{\Gamma(\kappa+1)} \sum_{\lambda_k < z}{}' \alpha_k \big(z - \lambda_k\big)^{\kappa}$$

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$$= z^{\frac{r+\kappa}{2}} \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{\frac{r+\kappa}{2}}} J_{r+\kappa}\big(2\sqrt{\mu_k z}\,\big) + \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\chi(w)\, z^{\rho+w}}{\Gamma(1+\kappa+w)}, w = s_k \right). \tag{8.11}$$

By **differencing** due to Landau [LanII] (cf. §6.1.1 and [ChN3]) applied to (8.11), the formula for the summatory function of the coefficients of the imaginary quadratic field is deduced.

Corollary 8.1

$$\sum_{k \le x}{}' F(k) = -\zeta_\Omega(0)x + \zeta_\Omega(0) + \sqrt{x} \sum_{k=1}^{\infty} \frac{F(k)}{\sqrt{k}}\, J_1\left(4\pi \sqrt{\frac{kx}{\sqrt{|\Delta|}}} \right) \tag{8.12}$$

by (3.151). The series on the right of (8.12) is convergent for all $x > 0$, boundedly convergent on any closed interval and uniformly convergent on any closed interval free from any jump discontinuities of the left side.

(8.12) was first deduced formally by Voronoĭ [Vor1] and rigorously proved by Hardy [HarCP]. Then it was generalized by Arnold Walfisz [AZW1], [AZW2] and further simplified by Anna Walfisz [AAW1].

Since a very wide class of zeta-functions (including those which are associated to cusp forms and positive definite quadratic forms) satisfies the function equation (8.1), Corollary 8.1 holds true for such zeta-functions. For the summatory function for a real quadratic field, cf. §9.1.2, especially (9.36).

8.3 The partial sum formula: $G^{2,0}_{2,2} \leftrightarrow G^{1,1}_{1,3}$

$$\begin{aligned} &\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s}\, G^{2,0}_{2,2}\left(z\lambda_k \,\middle|\, \begin{matrix} a, c \\ s, b \end{matrix} \right) \\ &= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}}\, G^{1,1}_{1,3}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-b \\ r-s, 1-a, 1-c \end{matrix} \right) \\ &\quad + \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\Gamma(a-s+w)}{\Gamma(b-s+w)\,\Gamma(c-s+w)}\, \chi(w)\, z^{s-w}, w = s_k \right). \end{aligned} \tag{8.13}$$

Equation (8.9) with $c = b$ amounts to (8.9), whence to Theorem 8.1, the Riesz sum formula. We remark that (8.13) with $c = s$ leads to another

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Riesz sum:

$$\sum_{k=1}^{\infty}\frac{\alpha_k}{\lambda_k^s}\,G_{1,1}^{1,0}\left(z\,\middle|\,\begin{matrix}a\\b\end{matrix}\right)=\sum_{k=1}^{\infty}\frac{\beta_k}{\mu_k^{r-s}}\,G_{1,3}^{1,1}\left(\frac{\mu_k}{z}\,\middle|\,\begin{matrix}1-b\\r-s,1-s,1-a\end{matrix}\right)$$
$$+\sum_{k=1}^{L}\operatorname{Res}\left(\frac{\Gamma(b-s+w)}{\Gamma(a-s+w)\,\Gamma(w)}\,\chi(w)\,z^{s-w},w=s_k\right).$$

Now, on appealing to (2.76) with $p=1,q=2$

$$G_{1,3}^{1,1}\left(z\,\middle|\,\begin{matrix}1-b\\r-s,1-s,1-a\end{matrix}\right)$$
$$=\frac{\Gamma(b+r-s)}{\Gamma(r)\,\Gamma(a+r-s)}z^{r-s}\,{}_1F_2\left(\begin{matrix}b+r-s\\r,a+r-s\end{matrix};-z\right),$$

and slightly changing notation, we obtain the Riesz sum formula

$$\frac{1}{\Gamma(\varkappa+1)}\sum_{\lambda_k\le z}{}'\frac{\alpha_k}{\lambda_k^{s-a}}(z-\lambda_k)^{\varkappa}\tag{8.14}$$
$$=\frac{\Gamma(a+r-s)\,z^{\varkappa+a+r-s}}{\Gamma(r)\Gamma(\varkappa+1+a+r-s)}\sum_{k=1}^{\infty}\beta_k\,{}_1F_2\left(\begin{matrix}a+r-s\\r,\varkappa+1+a+r-s\end{matrix};-z\mu_k\right)$$
$$+\sum_{k=1}^{L}\operatorname{Res}\left(\frac{\Gamma(a-s+w)}{\Gamma(\varkappa+1+a-s+w)\Gamma(w)}\,\chi(w)\,z^{\varkappa+a-s+w},w=s_k\right).$$

From (8.14) we may readily read off a partial sum or the Dirichlet polynomial for the zeta-function.

8.4 The Fourier-Bessel expansion: $G_{1,1}^{1,1}\leftrightarrow G_{0,2}^{2,0}$

This is expounded in more general form in Chapter 4.

8.5 The Ewald expansion: $G_{1,2}^{2,0}\leftrightarrow G_{1,2}^{1,1}$

This is expounded in more general form in Chapter 5. Cf. especially, Theorems 5.2 and 5.3. The latter implies the incomplete gamma expansion (Ewald expansion) from the confluent hypergeometric function expansion.

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8.6 The Bochner-Chandrasekharan formula: $\boldsymbol{H^{2,0}_{1,2} \leftrightarrow H^{1,1}_{1,2}}$

This has been expounded as §5.1.2.

8.7 The $\boldsymbol{G^{2,1}_{1,2} \leftrightarrow G^{2,1}_{1,2}}$ formula

This being a special case of (8.8), we find it more appropriate to state the counterpart of (8.6) in the case of the Riemann type functional equation, i.e. we assume that the gamma factor in (8.1) is $\Gamma\left(\frac{s}{2}\right)$

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s} H^{m+1,n}_{p,q+1}\left(z\lambda_k \,\middle|\, \begin{matrix} \{(1-a_j, A_j)\}_{j=1}^{n}, \{(a_j, A_j)\}_{j=n+1}^{p} \\ \left(\frac{s}{2}, \frac{1}{2}\right), \{(b_j, B_j)\}_{j=1}^{m}, \{(1-b_j, B_j)\}_{j=m+1}^{q} \end{matrix} \right) \tag{8.15}$$

$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}} H^{n+1,m}_{q,p+1}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} \{(1-b_j, B_j)\}_{j=1}^{m}, \{(b_j, B_j)\}_{j=m+1}^{q} \\ \left(\frac{r-s}{2}, \frac{1}{2}\right), \{(a_j, A_j)\}_{j=1}^{n}, \{(1-a_j, A_j)\}_{j=n+1}^{p} \end{matrix} \right)$$

$$+ \sum_{k=1}^{L} \operatorname{Res}\Big(\Gamma(w-s \,|\, \Delta)\, \chi(w)\, z^{s-w}, w = s_k \Big).$$

This leads to the confluent hypergeometrc expansion and thence to incomplete gamma expansion similar to the one in Chapter 5. It further boils down to the Ueno-Nishizawa formula [UN] ([Lav1]), which as has been enunciated in several papers of ours leads to the Hurwitz formula [Vista].

8.8 The second J-Bessel expansion: $\boldsymbol{G^{1,1}_{2,2} \leftrightarrow G^{2,0}_{1,3}}$

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s} G^{1,1}_{2,2}\left(z\lambda_k \,\middle|\, \begin{matrix} 1-b, c \\ s, 1-a \end{matrix} \right)$$

$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}} G^{2,0}_{1,3}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} a \\ r-s, b, 1-c \end{matrix} \right)$$

$$+ \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\Gamma(b+s-w)}{\Gamma(a+s-w)\,\Gamma(c-s+w)}\, \chi(w)\, z^{s-w}, w = s_k \right).$$

We do not have a closed form for $G^{2,0}_{1,3}$ in general and only partial results are known including (9.11), cf. Remark 9.1. Here we refer to the special case $a = r - s$, in which $G^{2,0}_{1,3}$ reduces to $G^{1,0}_{0,2}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} - \\ b, 1-c \end{matrix} \right)$, the J-Bessel

function by (2.110), which is also a consequence of (9.11).

On setting $z \leftarrow \frac{1}{z}$, we obtain

$$\begin{aligned}
&\frac{\Gamma(b+s)z^{-s}}{\Gamma(c-s)\Gamma(r)}\sum_{\lambda_k<z}\alpha_k\,{}_2F_1\left(\begin{matrix}b+s,1-c+s\\ r\end{matrix};\frac{\lambda_k}{z}\right) \qquad (8.16)\\
&+\frac{\Gamma(b+s)z^{b}}{\Gamma(-b-s+r)\Gamma(b+c)}\sum_{\lambda_k>z}\frac{\alpha_k}{\lambda_k^{s+b}}\,{}_2F_1\left(\begin{matrix}b+s,b+1-r+s\\ b+c\end{matrix};\frac{z}{\lambda_k}\right)\\
&=\sum_{k=1}^{\infty}\frac{\beta_k}{\mu_k^{r-s}}\sqrt{z\mu_k}^{\,1+b-c}J_{b+c-1}(2\sqrt{z\mu_k}\,)\\
&\quad+\sum_{k=1}^{L}\mathrm{Res}\left(\frac{\Gamma(b+s-w)}{\Gamma(r-w)\,\Gamma(c-s+w)}\chi(w)\,z^{w-s},w=s_k\right).
\end{aligned}$$

Here to the left-hand side, we applied the special case of (2.74):

$$\begin{aligned}
&G^{1,1}_{2,2}\left(z\,\middle|\,\begin{matrix}a_1,a_2\\ b_1,b_2\end{matrix}\right) \qquad (8.17)\\
&=\begin{cases}
\dfrac{\Gamma(1-a_1+b_1)}{\Gamma(a_2-b_1)\,\Gamma(1-b_2+b_1)}\,z^{b_1}\\
\qquad\qquad\times{}_2F_1\left(\begin{matrix}1-a_1+b_1,1-a_2+b_1\\ 1-b_2+b_1\end{matrix};z\right), & |z|<1\\
\dfrac{\Gamma(1-a_1+b_1)}{\Gamma(a_1-b_2)\,\Gamma(1-a_1+a_2)}\,z^{a_1-1}\\
\qquad\qquad\times{}_2F_1\left(\begin{matrix}1-a_1+b_1,1-a_1+b_2\\ 1-a_1+a_2\end{matrix};\dfrac{1}{z}\right), & |z|>1
\end{cases}
\end{aligned}$$

and for curiosity we note its special case.

$$G^{1,1}_{2,2}\left(z\,\middle|\,\begin{matrix}a,c\\ b,c\end{matrix}\right)=\begin{cases}\dfrac{\Gamma(1-a+b)}{\pi}\,\sin((c-b)\pi)\,z^b\,(1-z)^{a-b-1}, & |z|<1\\ \dfrac{\Gamma(1-a+b)}{\pi}\,\sin((a-c)\pi)\,z^b\,(z-1)^{a-b-1}, & |z|>1\end{cases}.$$

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8.9 The $H^{2,1}_{2,2} \leftrightarrow H^{2,1}_{1,3}$ formula

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s}\, H^{2,1}_{2,2}\left(z\lambda_k \,\middle|\, \begin{matrix} (1-b,B),(c,C) \\ (s,1),(a,A) \end{matrix} \right)$$
$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}}\, H^{2,1}_{1,3}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} (1-a,A) \\ (r-s,1),(b,B),(1-c,C) \end{matrix} \right)$$
$$+ \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\Gamma(a-A(s-w))\Gamma(b+B(s-w))}{\Gamma(c-C(s-w))}\, \chi(w)\, z^{s-w}, w = s_k \right).$$

We state its special case with $c = s, C = 1,\ A = B = 2, b = 0$, so that on the left we have $H^{1,1}_{1,1}\left(z\,\lambda_k \,\middle|\, \begin{matrix} (1,2) \\ (a,2) \end{matrix} \right)$, a reminiscent of the Fourier-Bessel expansion in Chapter 4.

$$\frac{\Gamma(2s)}{2} \sum_{k=1}^{\infty} \frac{\alpha_k}{\left(\sqrt{\lambda_k} + \sqrt{z}^{\,-1}\right)^{2s}} \tag{8.18}$$
$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}}\, H^{2,1}_{1,3}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} (1-a,2) \\ (r-s,1),(0,2),(1-s,1) \end{matrix} \right)$$
$$+ \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\Gamma(a-2s+2w))\Gamma(2s-2w)}{\Gamma(w)}\, \chi(w)\, z^{s-w}, w = s_k \right)$$

because

$$H^{1,1}_{1,1}\left(z \,\middle|\, \begin{matrix} (1,2) \\ (a,2) \end{matrix} \right) = \frac{1}{2}\, G^{1,1}_{1,1}\left(\sqrt{z} \,\middle|\, \begin{matrix} 1 \\ a \end{matrix} \right) = \frac{\Gamma(a)}{2} \frac{1}{\left(1 + \sqrt{z}^{\,-1}\right)^{a}}.$$

We state the results for $r = \frac{1}{2}$ with $a = 2s$ and $r = \frac{3}{2}$ with $a = 2s - 1$ for curiosity.

For the H-function on the right of (8.18), we apply the formula (2.11), which reads in the case $a = 2s$

$$H^{2,1}_{1,3}\left(z \,\middle|\, \begin{matrix} (1-2s,2) \\ (\frac{1}{2}-s,1),(0,2),(1-s,1) \end{matrix} \right)$$
$$= \frac{\Gamma(2s)}{2^{2-2s}\sqrt{\pi}} \Big(e^{-\frac{1-2s}{2}\pi\, i + 2\sqrt{z}\, i} \Gamma\big(1-2s, 2\sqrt{z}\, i\big) \tag{8.19}$$
$$+ e^{\frac{1-2s}{2}\pi\, i - 2\sqrt{z}\, i} \Gamma\big(1-2s, -2\sqrt{z}\, i\big) \Big).$$

Substituting (8.19) and making the change of variable $z \leftarrow \frac{1}{z^2}$, we obtain

$$\begin{aligned}
&\frac{\Gamma(2s)}{2}\sum_{k=1}^{\infty}\frac{\alpha_k}{\left(\sqrt{\lambda_k}+z\right)^{2s}}\\
&=\frac{\Gamma(2s)}{2^{2-2s}\sqrt{\pi}}\sum_{k=1}^{\infty}\frac{\beta_k}{\mu_k^{\frac{1}{2}-s}}\Big(e^{-\frac{1-2s}{2}\pi\, i+2z\sqrt{\mu_k}\, i}\,\Gamma(1-2s,2z\sqrt{\mu_k}\, i)\\
&\qquad\qquad + e^{\frac{1-2s}{2}\pi\, i-2z\sqrt{\mu_k}\, i}\,\Gamma(1-2s,-2z\sqrt{\mu_k}\, i)\Big)\\
&\quad+\sum_{k=1}^{L}\operatorname{Res}\left(\frac{\Gamma(2w)\Gamma(2s-2w)}{\Gamma(w)}\,\chi(w)\,z^{2w-2s},w=s_k\right).
\end{aligned}\tag{8.20}$$

In exactly the same way, (8.18) with $r=\frac{3}{2}, a=2s-1$ amounts to

$$\begin{aligned}
&\frac{\Gamma(2s-1)}{2}\sum_{k=1}^{\infty}\frac{\alpha_k}{\sqrt{\lambda_k}\left(\sqrt{\lambda_k}+z\right)^{2s-1}}\\
&=\frac{\Gamma(2s-1)}{2^{3-2s}\sqrt{\pi}}\sum_{k=1}^{\infty}\frac{\beta_k}{\mu_k^{\frac{3}{2}-s}}\Big(ie^{-(1-s)\pi i+2z\sqrt{\mu_k}\, i}\,\Gamma(2-2s,2z\sqrt{\mu_k}\, i)\\
&\qquad\qquad - ie^{(1-s)\pi i-2z\sqrt{\mu_k}\, i}\,\Gamma(2-2s,-2z\sqrt{\mu_k}\, i)\Big)\\
&\quad+\sum_{k=1}^{L}\operatorname{Res}\left(\frac{\Gamma(2w-1)\Gamma(2s-2w)}{\Gamma(w)}\,\chi(w)\,z^{2w-2s},w=s_k\right).
\end{aligned}\tag{8.21}$$

8.10 The second K-Bessel expansion: $G^{3,0}_{1,3} \leftrightarrow G^{1,2}_{2,2}$

$$\begin{aligned}
&\sum_{k=1}^{\infty}\frac{\alpha_k}{\lambda_k^s}\,G^{3,0}_{1,3}\left(z\lambda_k\,\middle|\,\begin{matrix}c\\ s,a,b\end{matrix}\right)\\
&=\sum_{k=1}^{\infty}\frac{\beta_k}{\mu_k^{r-s}}\,G^{1,2}_{2,2}\left(\frac{\mu_k}{z}\,\middle|\,\begin{matrix}1-a,1-b\\ r-s,1-c\end{matrix}\right)\\
&\quad+\sum_{k=1}^{L}\operatorname{Res}\left(\frac{\Gamma(a-s+w)\,\Gamma(b-s+w)}{\Gamma(c-s+w)}\,\chi(w)\,z^{s-w},w=s_k\right),
\end{aligned}$$

which, for $b=c=s$, reduces to

$$G^{2,0}_{0,2}\left(z\lambda_k\,\middle|\,\begin{matrix}-\\ a,b\end{matrix}\right)\leftrightarrow G^{1,1}_{1,1}\left(\frac{\mu_k}{z}\,\middle|\,\begin{matrix}1-a\\ r-s\end{matrix}\right),$$

the Fourier-Bessel expansion in Chapter 5.

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For $c = s$, the right-hand side member becomes the K-Bessel function by (2.109) and we have

$$2\sum_{k=1}^{\infty}\frac{\alpha_k}{\lambda_k^s}\sqrt{z\lambda_k}^{\,a+b}K_{b-a}\Big(2\sqrt{z\lambda_k}\Big) \tag{8.22}$$
$$= \frac{\Gamma(a+r-s)\,\Gamma(b+r-s)}{\Gamma(r)}\,z^{s-r}\sum_{k=1}^{\infty}\beta_k\,{}_2F_1\left(\begin{matrix}a+r-s, b+r-s\\ r\end{matrix};-\frac{\mu_k}{z}\right)$$
$$+\sum_{k=1}^{L}\mathrm{Res}\left(\frac{\Gamma(a-s+w)\,\Gamma(b-s+w)}{\Gamma(w)}\,\chi(w)\,z^{s-w}, w=s_k\right),$$

on using (2.76).

8.11 The $G^{3,1}_{1,3} \leftrightarrow G^{2,2}_{2,2}$ formula

$$\sum_{k=1}^{\infty}\frac{\alpha_k}{\lambda_k^s}\,G^{3,1}_{1,3}\left(z\lambda_k\,\middle|\,\begin{matrix}1\\ s,a,b\end{matrix}\right)$$
$$=\sum_{k=1}^{\infty}\frac{\beta_k}{\mu_k^{r-s}}\,G^{2,2}_{2,2}\left(\frac{\mu_k}{z}\,\middle|\,\begin{matrix}1-a,1-b\\ 0,r-s\end{matrix}\right)$$
$$+\sum_{k=1}^{L}\mathrm{Res}\Big(\Gamma(a-s+w)\,\Gamma(b-s+w)\,\Gamma(s-w)\,\chi(w)\,z^{s-w}, w=s_k\Big),$$

which leads to a formula involving ${}_1F_2$ and ${}_2F_2$ by (2.74).

8.12 The $G^{2,1}_{2,3} \leftrightarrow G^{2,1}_{2,3}$ formula

This is another special case of (8.8):

$$\sum_{k=1}^{\infty}\frac{\alpha_k}{\lambda_k^s}\,G^{2,1}_{2,3}\left(z\lambda_k\,\middle|\,\begin{matrix}1-a+c,b\\ s,a,1-b+c\end{matrix}\right)$$
$$=z^c\sum_{k=1}^{\infty}\frac{\beta_k}{\mu^{r+c-s}}\,G^{2,1}_{2,3}\left(\frac{\mu_k}{z}\,\middle|\,\begin{matrix}1-a+c,b\\ r-s+c,a,1-b+c\end{matrix}\right) \tag{8.23}$$
$$+\sum_{k=1}^{L}\mathrm{Res}\left(\frac{\Gamma(a-s+w)\Gamma(a-c+s-w)}{\Gamma(b-s+w)\Gamma(b-c+s-w)}\,\chi(w)\,z^{s-w}, w=s_k\right),$$

which leads to a formula involving ${}_2F_2$ by (2.74).

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8.13 The $G^{3,0}_{2,3} \leftrightarrow G^{1,2}_{2,3}$ formula

We have

$$\begin{aligned}
&\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s}\, G^{3.0}_{2,3}\left(z\lambda_k \,\middle|\, \begin{matrix} b_1, b_2 \\ a, a_1, a_2 \end{matrix} \right) \\
&= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu^{r-s}}\, G^{1,2}_{2,3}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-a_1, 1-a_2 \\ r-s, 1-b_1, 1-b_2 \end{matrix} \right) \\
&\quad + \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\Gamma(a_1 - s + w)\,\Gamma(a_2 - s + w)}{\Gamma(b_1 - s + w)\,\Gamma(b_2 - s + w)}\, \chi(w)\, z^{s-w}, w = s_k \right),
\end{aligned} \tag{8.24}$$

which leads to a formula involving ${}_2F_2$ by (2.74).

8.14 The $G^{3,1}_{2,3} \leftrightarrow G^{2,2}_{2,3}$ formula

We have

$$\begin{aligned}
&\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s}\, G^{3,1}_{2,3}\left(z\lambda_k \,\middle|\, \begin{matrix} 1, c \\ s, a, b \end{matrix} \right) \\
&= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu^{r-s}}\, G^{2,2}_{2,3}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-a, 1-b \\ 0, r-s, 1-c \end{matrix} \right) \\
&\quad + \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\Gamma(a - s + w)\,\Gamma(b - s + w)\,\Gamma(s - w)}{\Gamma(c - s + w)}\, \chi(w)\, z^{s-w}, w = s_k \right),
\end{aligned} \tag{8.25}$$

which by (2.74) leads to the ${}_2F_2$ formula adopted in the comic [KH]:

$$\begin{aligned}
\sum_{k=1}^{\infty} \alpha_k \Bigg\{ & \frac{\Gamma(s-a)\Gamma(b-a)\Gamma(a)}{\Gamma(c-a)} \frac{z^a}{\lambda_k^{s-a}}\, {}_2F_2\left(\begin{matrix} a, 1+a-c \\ 1+a-s, 1+a-b \end{matrix} ; z\lambda_k \right) \\
& + \frac{\Gamma(s-b)\Gamma(a-b)\Gamma(b)}{\Gamma(c-b)} \frac{z^b}{\lambda_k^{s-b}}\, {}_2F_2\left(\begin{matrix} b, 1+b-c \\ 1+b-s, 1+b-a \end{matrix} ; z\lambda_k \right) \\
& + \frac{\Gamma(a-s)\Gamma(b-s)\Gamma(s)}{\Gamma(c-s)}\, z^s\, {}_2F_2\left(\begin{matrix} s, 1+s-c \\ 1+s-a, 1+s-b \end{matrix} ; z\lambda_k \right) \Bigg\}
\end{aligned}$$

$$= \sum_{k=1}^{\infty} \beta_k \left\{ \frac{\Gamma(r-s)\Gamma(a)\Gamma(b)}{\Gamma(c)} \frac{1}{\mu_k^{r-s}} \, {}_2F_2\left(\begin{matrix} a, b \\ 1+s-r, c \end{matrix} ; \frac{\mu_k}{z} \right) \right. \tag{8.26}$$

$$+ \frac{\Gamma(s-r)\Gamma(a+r-s)\Gamma(b+r-s)}{\Gamma(c+r-s)} \frac{1}{z^{r-s}}$$

$$\left. \times \, {}_2F_2\left(\begin{matrix} a+r-s, b+r-s \\ 1+r-s, c+r-s \end{matrix} ; \frac{\mu_k}{z} \right) \right\}$$

$$+ \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\Gamma(a-s+w)\,\Gamma(b-s+w)\,\Gamma(s-w)}{\Gamma(c-s+w)} \, \chi(w)\, z^{s-w}, w = s_k \right).$$

The left-hand side member, in the case $c = s$, reduces to ${}_1F_1$.

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s} \, G_{1,2}^{2,1}\left(z\lambda_k \,\middle|\, \begin{matrix} 1 \\ a, b \end{matrix} \right)$$

$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}} \, G_{2,3}^{2,2}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-a, 1-b \\ 0, r-s, 1-s \end{matrix} \right)$$

$$+ \sum_{k=1}^{L} \operatorname{Res}\left(\frac{\Gamma(a-s+w)\,\Gamma(b-s+w)\,\Gamma(s-w)}{\Gamma(w)} \, \chi(w)\, z^{s-w}, w = s_k \right).$$

8.15 The $G_{2,3}^{3,2} \leftrightarrow G_{2,3}^{3,2}$ formula

This is another example of the bilateral formula.

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s} \, G_{2,3}^{3,2}\left(z\lambda_k \,\middle|\, \begin{matrix} 1-a+c, 1-b+c \\ s, a, b \end{matrix} \right)$$

$$= z^c \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r+c-s}} \, G_{2,3}^{3,2}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-a+c, 1-b+c \\ r-s+c, a, b \end{matrix} \right) \tag{8.27}$$

$$+ \sum_{k=1}^{L} \operatorname{Res}\Big(\Gamma(a-s+w)\,\Gamma(b-s+w)\,\Gamma(a-c+s-w)\,\Gamma(b-c+s-w)$$

$$\chi(w)\, z^{s-w}, w = s_k \Big),$$

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which leads to a formula similar to but more general than (8.26). We state it for curiosity's sake.

$$\begin{aligned}
\sum_{k=1}^{\infty}\alpha_k\Bigg\{&\Gamma(a-s)\Gamma(b-s)\Gamma(a-c+s)\Gamma(b-c+s)\\
&\qquad\times z^s\,{}_2F_2\left(\begin{matrix}a-c+s,b-c+s\\1-a+s,1-b+s\end{matrix};-z\lambda_k\right)\\
&+\Gamma(s-a)\Gamma(b-a)\Gamma(2a-c)\Gamma(a+b-c)\\
&\qquad\times\frac{z^a}{\lambda_k^{s-a}}\,{}_2F_2\left(\begin{matrix}2a-c,a+b-c\\1+a-s,1+a-b\end{matrix};-z\lambda_k\right)\\
&+\Gamma(s-b)\Gamma(a-b)\Gamma(2b-c)\Gamma(a+b-c)\\
&\qquad\times\frac{z^b}{\lambda_k^{s-b}}\,{}_2F_2\left(\begin{matrix}2b-c,a+b-c\\1+b-s,1+b-a\end{matrix};-z\lambda_k\right)\Bigg\}\\
=\sum_{k=1}^{\infty}\beta_k\Bigg\{&\Gamma(a-c-r+s)\Gamma(b-c-r+s)\Gamma(a+r-s)\Gamma(b+r-s)\\
&\qquad\times\frac{1}{z^{r-s}}\,{}_2F_2\left(\begin{matrix}a+r-s,b+r-s\\1-a+c+r-s,1-b+c+r-s\end{matrix};-\frac{\mu_k}{z}\right)\\
&+\Gamma(r-s-a+c)\Gamma(b-a)\Gamma(2a-c)\Gamma(a+b-c)\\
&\qquad\times\frac{z^{c-a}}{\mu_k^{r-s+c-a}}\,{}_2F_2\left(\begin{matrix}2a-c,a+b-c\\1+a-c-r+s,1+a-b\end{matrix};-\frac{\mu_k}{z}\right)\\
&+\Gamma(r-s-b+c)\Gamma(a-b)\Gamma(2b-c)\Gamma(a+b-c)\\
&\qquad\times\frac{z^{c-b}}{\mu_k^{r-s+c-b}}\,{}_2F_2\left(\begin{matrix}2b-c,a+b-c\\1+b-c-r+s,1+b-a\end{matrix};-\frac{\mu_k}{z}\right)\Bigg\}\\
+\sum_{k=1}^{L}\operatorname{Res}\Big(&\Gamma(a-s+w)\,\Gamma(b-s+w)\\
&\qquad\times\Gamma(a-c+s-w)\,\Gamma(b-c+s-w)\,\chi(w)\,z^{s-w},\\
&\qquad\qquad w=s_k\Big).\qquad(8.28)
\end{aligned}$$

8.16 The $G^{p+1,0}_{1,p+1} \leftrightarrow G^{1,p}_{p,2}$ formula

This has some relevance to §3.1 and §9.2.

$$\sum_{k=1}^{\infty}\frac{\alpha_k}{\lambda_k^s}\,G^{p+1,0}_{1,p+1}\left(z\lambda_k\,\middle|\,\begin{matrix}b\\s,a_1,\dots,a_p\end{matrix}\right)$$

$$= \sum_{k=1}^{\infty} \frac{\beta_k}{\mu_k^{r-s}} G_{p,2}^{1,p}\left(\frac{\mu_k}{z} \,\middle|\, \begin{matrix} 1-a_1, \ldots, 1-a_p \\ r-s, 1-b \end{matrix} \right)$$

$$+ \sum_{k=1}^{L} \operatorname{Res} \left(\frac{\prod_{j=1}^{p} \Gamma(a_j - s + w)}{\Gamma(b - s + w)} \chi(w)\, z^{s-w}, w = s_k \right), \tag{8.29}$$

which reduces, in the special case $b = s$, to

$$\sum_{k=1}^{\infty} \frac{\alpha_k}{\lambda_k^s} V(z\lambda_k; a_1, \cdots, a_p) \tag{8.30}$$

$$= \frac{\prod_{j=1}^{p} \Gamma(a_j + r - s)}{\Gamma(r)} z^{s-r}$$

$$\times \sum_{k=1}^{\infty} \beta_k \, {}_pF_1\left(\begin{matrix} a_1 + r - s, \ldots, a_p + r - s \\ r \end{matrix} ; -\frac{\mu_k}{z} \right)$$

$$+ \sum_{k=1}^{L} \operatorname{Res} \left(\frac{\prod_{j=1}^{p} \Gamma(a_j - s + w)}{\Gamma(w)} \chi(w)\, z^{s-w}, w = s_k \right),$$

where V is Steen's function defined by (3.9) and we applied (2.76) on the right.

In the special case $a_1 = 0$, $a_2 = \frac{1}{p}, \ldots, a_p = \frac{p-1}{p}$ we have the formula

$$G_{0,p}^{p,0}\left(z\lambda_k \,\middle|\, \begin{matrix} - \\ 0, \frac{1}{p}, \ldots, \frac{p-1}{p} \end{matrix} \right) = \frac{\sqrt{2\pi}^{\,p-1}}{\sqrt{p}} e^{-p \sqrt[p]{z\lambda_k}} \tag{8.31}$$

and so the left-hand side member of (8.30) reduces to the exponential function.

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